

① REQUIREMENTS — canonical enterprise-Q&A RAG workload

~2,000 users · ~10 rps avg / 40 rps peak
9.2K input + 300 output tokens per request (TTFT $\leq$ 1.2 s, TPOT ~25 ms)
Quality class: ~70B answer quality

② CONSTRAINTS — hard gates (constraint-first synthesis)

MEMORY total residency $\leq$ host aggregate (weights + KV)
LATENCY TTFT $\leq$ 1.2 s (p95 $\leq$ 2 s) · TPOT ~25 ms
ECONOMICS ~15K/moper — host ceiling (~14.6K/mo per 8xH100)

③ GENERATE CANDIDATES (each must fit memory, latency, economics)

(a) 8xH100 + prefix cache (b) P/D-disaggregated 2-pool
weights 140 GB + KV 24 GB = 164 GB weights 140 GB + KV 24 GB = 164 GB/pool
(c) KV-quantized 7B — weights 14 GB + KV 6 GB = 20 GB

④ TEST AGAINST memory · latency · economics

(a) 8xH100 + prefix cache

MEM 164 $\leq$ 640 GB ✓ | TTFT ✓ in SLO | 1 host $\approx$ \$14.6K/mo ✓
→ memory/latency OK, but prefill ~19.8K $\ll$ ~92K needed
✗ capacity (needs ~5 hosts)

(b) P/D-disaggregated 2-pool

MEM 164 $\leq$ 640 GB/pool ✓ | TTFT ✓ phase-split | 2 hosts + fabric (~2x capex)
→ survives as the at-scale answer (prefill pool ~19.8K)

(c) KV-quantized 7B

MEM 20 $\leq$ 80 GB ✓ | ✗ QUALITY — provisional | cheapest (1 H100)
→ throughput high, but fails quality gate ✗ EXCLUDED

⑤ OUTCOME — SURVIVORS

(a) per-request-latency baseline
(b) at-scale P/D answer — carried to Ch.13-14

✗ REJECTED

(c) fails quality gate
→ excluded from evaluation